

Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
University of Pisa

Project Report

Academic Year 20XX–20XX

Project Title: [Insert Project Title Here]

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Submission date: DD/MM/YYYY

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How to Use This Template

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Before submitting, make sure to:

- Delete this page entirely.
- Remove all `instruction` environments (the shaded blue boxes). They are guidance for the authors and must not appear in the final submission.
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- Keep section titles, font (Helvetica 12 pt), and margins unchanged. You may add subsections.
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About the instruction boxes: The shaded boxes with a blue left border are guidance on what to write in each section. Read them, then delete them before submission.

About Table 1: Table 1 is a snapshot of your problem and dataset. Adapt its rows to your task: add rows if needed, remove those that do not apply. Keep cell content concise — one or two lines per cell at most.

About figures: Every placeholder marks the position of a figure and suggests a chart type. You are free to choose the most appropriate visualization. What matters is that every figure is numbered, has a descriptive caption, and is explicitly referenced in the text before it appears.

Abstract

Concise summary of the entire project: max 250 words. Cover: (1) problem addressed, (2) dataset and task type, (3) methods used, (4) main results, (5) key conclusions. Write in the third person. Do not cite references here.

[Write abstract here — max 250 words]

1. Introduction

Suggested length: ~1 page.

Briefly describe the application domain and the problem. Explain why it matters (social, scientific, economic relevance). Cite 2–3 references to support your claims. End with a numbered list of project objectives.

[Provide background on the problem and its relevance. Support claims with references [? ?].]

This project aims to:

1. [Objective 1: e.g., develop a predictive model for...]
2. [Objective 2: e.g., compare the performance of multiple model families...]
3. [Objective 3: e.g., provide global and local model explanations using...]

References

- [1] Author, A., Author, B. (Year). Title. *Journal/Conference*, Vol(No), pp. XXX–XXX.
- [2] Author, C. (Year). *Title of the book*. Publisher.

2. Problem

Suggested length: ~2 pages.

Formally describe the problem. **Use short sentences that go straight to the point.** Limit the use of adverbs and useless words that only make the text hard to understand without adding any information. Use the **active form** with the subject at the beginning. Examples of convoluted sentences to avoid:

- 1) Figure 1 ~~contains a plot that was specifically generated to leverage a differentiated series of views to comprehensively visualize the intricate behavior of~~ **shows** the feature values;
- 2) These particular and specific feature values, in some peculiar cases, have the potential to **can** modify the behavior
- 3) This result ~~is not only a very interesting indicator to perfectly and completely describe the variations of the labels that the classifier associates with certain samples, but it is also a powerful indicator of~~ **quantifies** the mistakes the classifier makes.
- 4) Our techniques ~~are able to completely~~ eliminate overfitting.

In this section, specify the input features (what they represent, their type, how they were obtained, and why they were selected), the target variable, and the ML task type. If features come from previous works, cite the sources. Table ?? is a snapshot; adapt its rows to your problem. Keep cells concise: one or two lines at most.

Table ?? summarizes the problem and dataset.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	e.g., Binary classification
Dataset	e.g., UCI Heart Disease — 303 samples, 13 features
Target variable	e.g., Presence of heart disease (0 / 1)
Features	e.g., 13 features: 6 numerical, 7 categorical — see Section ??
Missing values	e.g., 1.3% overall; cholesterol: 4.6% (class 0: 4.1%, class 1: 5.1%)
Class distribution	e.g., Class 0: 54%, Class 1: 46% — approximately balanced
Evaluation metric(s)	e.g., F1-score (macro), ROC-AUC
Train / Val / Test	e.g., 70% / 10% / 20% — stratified

2.1 Exploratory Data Analysis

Report your exploratory analysis: feature distributions, class balance (or target distribution for regression/forecasting), correlations, and missing values. For missing values, report the percentage overall and per class. Use figures with clear captions, refer to them in the text, and number all figures. Do not discuss here how missing values or class imbalance were handled: this belongs to Section ??.

[Describe distributions, correlations, target distribution, and missing values. Refer to Figure ?? and Figure ??.]

[YOUR FIGURE HERE]

Replace with your figure. Suggested: histograms or box plots of feature distributions grouped by class label (classification) or colored by cluster (clustering).

Figure 1. Feature distributions by class label.

2.2 Feature Description

For each feature, specify: name, type (numerical/categorical), meaning in the domain, how it was obtained or computed, and why it is expected to be informative. If the feature set comes from the literature, a citation is sufficient. If features were engineered, describe the process.

[Describe and justify each feature. Refer to Table ?? as needed.]

[YOUR FIGURE HERE]

Replace with your figure. Suggested: Pearson correlation heatmap or scatter matrix of the most relevant features.

Figure 2. Correlation heatmap of numerical features.

3. Pipeline and Experiments

Suggested length: ~4 pages.

Describe the full ML pipeline from preprocessing to final model output. Justify every decision. Explicitly address: (1) how data leakage was avoided, (2) how missing values were imputed, (3) how class imbalance was handled (if applicable), (4) how the model family was selected, (5) how hyperparameters were optimized, (6) how generalization was assessed, and (7) how the final model was trained. Do not include code snippets; reference line numbers when describing key steps. Use numbered figures with captions and refer to them in the text.

3.1 Preprocessing

Describe normalization, encoding, imputation strategy, and any other data transformation. Explain why each step was performed and where in the pipeline it was applied to prevent data leakage. Reference relevant code lines.

[Describe preprocessing steps. Justify each choice. Reference code lines where relevant.]

3.2 Feature Selection and Engineering

Describe feature selection or engineering steps. Justify choices. Reference Figure ?? and discuss which features were retained and why.

[Describe feature selection/engineering. Refer to Figure ??.]

[YOUR FIGURE HERE]

Replace with your figure. Suggested: feature importance ranking (permutation importance, mean |SHAP|, mutual information, etc.).

Figure 3. Feature importance scores.

3.3 Model Selection and Cross-Validation

Explain which model families were considered and why. Describe the cross-validation strategy (e.g., nested CV, stratified k -fold), how you applied it, and why. State explicitly how data leakage was prevented (all transformations fitted inside CV folds). Describe how class imbalance was handled (e.g., SMOTE inside folds, class weights). Refer to Table ?? and Figure ??.

[Describe model selection rationale and CV strategy. Refer to Table ?? and Figure ??.]

Table 2. Comparison of candidate models.

Model	HP tuning	CV strategy	Val. metric	Test metric
Baseline (e.g., DT)	None	5-fold stratified	—	—
Model A (e.g., RF)	Grid Search	5-fold stratified	—	—
Model B (e.g., SVM)	Random Search	5-fold stratified	—	—
Model C (e.g., XGB)	Bayesian Opt.	5-fold stratified	—	—

[YOUR FIGURE HERE]

Replace with your figure. Suggested: box plot of cross-validated scores across candidate models.

Figure 4. Cross-validated scores by model.

3.4 Hyperparameter Optimization

Describe the search strategy (Grid Search, Random Search, or Bayesian Optimization), the search space for each hyperparameter, and the metric used to select the best configuration. Report the best values found. Refer to Figure ??.

[Describe hyperparameter optimization. Report best configuration. Refer to Figure ??.]

[YOUR FIGURE HERE]

Replace with your figure. Suggested: validation curve (metric vs. hyperparameter value) or 2D search grid heatmap.

Figure 5. Hyperparameter optimization results.

3.5 Statistical Tests

Describe statistical tests used to compare models (e.g., Wilcoxon signed-rank test, McNemar's test). Report test statistics and p -values. State your conclusions about statistical significance.

[Report statistical tests, test statistics, p -values, and conclusions.]

3.6 Final Model Training

Describe how the final model was trained after hyperparameter selection (e.g., retrained on the full train + validation set). State how generalization to unseen data was assessed.

[Describe final training procedure and generalization assessment.]

4. Results and Discussion

Suggested length: ~2.5 pages.

Do not list results mechanically. Discuss them critically and in relation to the problem domain. Explain what errors mean in practice: what are the consequences of a false positive or a false negative in this specific application? If prior work tackled the same task with different results, formulate motivated hypotheses about the causes (dataset size, feature set, evaluation protocol, data leakage, etc.). Use appropriate visualizations. Number all figures and refer to them in the text.

4.1 Performance Evaluation

Report test-set performance using appropriate metrics. Favor visual representations. Discuss what the numbers mean in context — not just whether they are high or low, but what types of errors the model makes and why they matter in this domain. Refer to Figure ?? and Figure ??.

[Discuss test-set performance. Refer to Figure ?? and Figure ??.]

[YOUR FIGURE HERE]

Replace with your figure. Classification: confusion matrix. Regression: predicted vs. actual scatter plot. Forecasting: actual vs. predicted time series. Clustering: 2D projection (PCA/t-SNE) with cluster labels.

Figure 6. Primary performance result on the test set.

[YOUR FIGURE HERE]

Replace with your figure. Classification: ROC or Precision-Recall curve. Regression: residual plot or error histogram. Forecasting: residual plot or error over time. Clustering: silhouette plot or elbow / gap-statistic curve.

Figure 7. Secondary performance metric on the test set.

4.2 Model Explainability

Apply at least one global technique (e.g., permutation importance, SHAP summary plot) and one local technique (e.g., LIME, SHAP force or waterfall plot). Discuss whether the explanations are consistent with domain knowledge. Use explainability to support or challenge conclusions drawn from performance metrics. Refer to Figure ?? and Figure ??.

[Describe and interpret global and local explanations. Refer to Figure ?? and Figure ??.]

[YOUR FIGURE HERE]

Replace with your figure. Suggested: mean |SHAP| bar chart or permutation importance plot — global explainability.

Figure 8. Global explainability: mean |SHAP| values.

[YOUR FIGURE HERE]

Replace with your figure. Suggested: SHAP waterfall or force plot for a representative sample — local explainability.

Figure 9. Local explainability: SHAP plot for a representative sample.

4.3 Error Analysis

Identify patterns in misclassified or mispredicted samples. Characterize their feature profiles. Use explainability outputs to reason about why the model fails on these cases. Connect the analysis to the domain: are these errors understandable? Are there feature combinations that systematically mislead the model? This section should reflect a genuine attempt to understand failure modes, not a generic acknowledgment that errors exist.

[Analyze error patterns. Use feature profiles and explainability outputs to reason about causes.]

5. Conclusions

Suggested length: ~0.5 pages.

Summarize the main findings and state whether each project objective (listed in Section 1) was met. Acknowledge limitations honestly. Suggest possible directions for future work. Keep this section short.

[Summarize key results, address each objective, discuss limitations, suggest future work.]

References

- [1] Author, A., Author, B. (Year). Title of the paper. *Journal / Conference*, Vol(No), pp. XXX–XXX.
- [2] Author, C. (Year). *Title of the book*. Publisher.